

(Task 1 of 10) In this tutorial, we will learn even more about definitions.

0

(Task 2 of 10) What is the result of running this program?

Python | Python [Run 

Pseudo

```
def f(x):          fun f(x):
    return 3       return 3
print(f(7 / 0))    end
                  print(f(7 / 0))
```

⚠ JavaScript behaves differently.

error

2

You predicted the output correctly 🎉🎉🎉

3

Function calls bind their formal parameters (in this case, there is one formal parameter, `x`) to the values of actual parameters (in this case, there is one actual parameter, `7 / 0`). The program errors when it tries to evaluate `7 / 0`.

Click [here](#) to run this program in the Stacker.

(Task 3 of 10) What is the result of running this program?

Python | Python [Run 

Pseudo

```
x = 12 / 0    let x = 12 / 0
print(3)      print(3)
```

⚠ JavaScript behaves differently.

error

5

You predicted the output correctly 🎉🎉🎉

6

When you define a variable (in this case, `x`), you have to bind it to a value, no matter whether or not you need the value of that variable later in the program. The program errors when it tries to evaluate `12 / 0`.

Click [here](#) to run this program in the Stacker.

Python | 

(Task 4 of 10) What is the result of running this program?

Python [Run]

Pseudo

<code>x = y + 1</code>	<code>let x = y + 1</code>
<code>y = 2</code>	<code>let y = 2</code>
<code>print(x)</code>	<code>print(x)</code>
<code>print(y)</code>	<code>print(y)</code>

error

8

Please briefly explain why you think the answer is error.

9

invalid def for y

10

You predicted the output correctly 🎉🎉🎉

11

The first definition tries to bind `x` to the value of `y + 1`. To evaluate `y + 1`, we need the value of `y`. But `y` is not bound to a value at that moment.

Click [here](#) to run this program in the Stacker.

(Task 5 of 10) In what order are definitions and expressions evaluated?

12

sequentially

13

(Task 6 of 10) - Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

14

- Every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

Any feedback regarding these statements? Feel free to skip this question.

15

cool stuff

16

Thank you!

17

(Task 7 of 10) Please scroll back and select 1-3 programs that together make these points. 18

- Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

You don't need to select all such programs.

(You selected 3 programs) 19

Okay. How do these programs (1,4,7) support the point? 20

all examples define variables invalidly or out of order 21

(Task 8 of 10) Please scroll back and select 1-3 programs that together make these points. 22

- Every block evaluates in top-to-bottom, left-to-right order.

You don't need to select all such programs.

(You selected 1 programs) 23

Okay. How does this program (4) support the point? 24

all programming langs evaluate sequentially as such. 25

(Task 9 of 10) Here is a program that confused many students

Python | Scala 3

```
Python [Run]
def addy(x):
    return x + y
s = addy(1)
y = 2
print(s)

Scala 3
def addy(x : Int) =
    x + y
val s = addy(1)
val y = 2
println(s)
```

Please

- Run this program in the stacker by clicking one of the [Run] buttons above;
- The stacker would show how this program produces its result(s);
- Keep clicking **Next** until you reach a configuration that you find particularly

helpful;

4. Click  Share This Configuration to get a link to your configuration;

5. Submit your link below;

`https://smol-tutor.xyz/stacker/?syntax=Python&inputSyntax=Lispy&randomSeed=smol-tutor&hole=%E2%80%A2&nNext=1&program=%0A%28deffun+%28addy+x%29%0A++%28%2B+x+y%29%29%0A%28defvar+s+%28addy+1%29%29%0A%28defvar+y+2%29%0A%0As%0A&readOnlyMode=` ²⁷

(Task 10 of 10) Please write a couple of sentences to explain how your configuration explains the result(s) of the program. ²⁸

The blue frame attempts to evaluate $x + y$ when y has not been defined. Therefore, it will error because y had not previously been defined in the call stack globally ²⁹

Let's review what we have learned in this tutorial. ³⁰

- Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.
- Every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

You have finished this tutorial 🎉🎉🎉

Please  the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

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